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# Sagnac effect in an off-center rotating ring frame of reference

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**Abstract**

Interference resides deeply in our understanding of the wave properties of light. In this paper, the century famous Sagnac effect is demonstrated to be independent of the rotation axis position, using a rotating ring optical fiber in a straightforward laboratory experiment. A simple theoretical explanation for this result is given for any arbitrary closed loop interferometer. The level of this discussion should be suitable for undergraduate physics or engineering courses where electromagnetic theory and optics are discussed. The experiment described utilizes basic and important aspects in modern optics which every science student should acquire.

Keywords: interference, light, Sagnac effect, rotating ring optical fiber, closed loop interferometer

(Some figures may appear in colour only in the online journal)

## 1. Introduction

Interference phenomena are central in many fields of physics such as electromagnetic theory of light and quantum mechanics. The Sagnac effect is a famous, century old experiment, where a beam of light is split and the two outcoming beams counter propagate along a closed loop and interfere [1, 2]. The central idea is that when the resulting setup is rotated, a path difference between the beams is created, thus changing the interference pattern. The path difference is linearly proportional to the rotation frequency of the system. This famous effect is manifested in various current research fields including atom interferometry [3–5], astronomy [6], general relativity effects such as Lense–Thirring precession [7, 8] and presently even is being explored as a possible advanced detection method for gravitational waves [9, 10].

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Another famous historical application of this effect was demonstrated by Michelson and Gale in 1926 to confirm the rotation frequency of the Earth [11]. However, the most common employment in current technology, is in the form of optical gyroscopes, which employ this effect for navigation by integrating the rotation frequency [12]. These are extremely sensitive to rotations, which need to be accounted for if a guidance system is to return correct results. For example, the global positioning system needs to take the rotation of Earth into account in the procedures of using radio signals to synchronize clocks [13].

The Sagnac effect manifests itself mostly in an experimental setup called a ring interferometer. A beam of light is split and the two beams are made to follow, in opposite directions, a common trajectory that constitutes a loop that must enclose an area. The light that exits the loop is guided in such a way that an interference pattern is obtained. The position of the interference fringes is dependent on the angular velocity of the setup. This arrangement is called a Sagnac interferometer.

This type of interferometer is also known as a passive ring interferometer, which uses light entering the setup from outside the ring. The interference pattern that is obtained is a fringe pattern, and what is measured is a phase shift. Another important form is the active ring interferometer (often referred to as a ring laser) which is self–contained in the sense that the light is generated and sustained by incorporating laser excitation in the path of the light [14, 15]. In recent years, large ring lasers have been successfully employed for geodesic measurements [16] and rotational seismology [17].

The rich history and current important applications of the Sagnac effect present a vibrant and beautiful ground for a student experiment demonstration of interference, which is one of the most important and fundamental phenomena in science. The relative simplicity of the theory and experiment offers undergraduate students of science and engineering an elementary implementation of basic aspects in physics and optics. In this paper, a student–laboratory Sagnac interferometer is described, utilizing an optical fiber ring (OFR), which is used to explore the dependence of the Sagnac effect on the rotation axis position of the system relative to the principal axis of the fiber ring.

## 2. Theory

The theoretical description of the Sagnac effect has been well known for over a century. Essential in it is the fact that the two beams propagate in a enclosed loop and form an interference pattern. Therefore, the important parameters are the angular velocity of the interferometer and the area circumnavigated by the beams. For a closed loop in a rotating frame of reference, where the axis of rotation is within the loop, the general expression for the change in phase ($\Delta\phi$) of the interference pattern is [18, 19],

$$\Delta\phi = \frac{8\pi N\vec{A}\cdot\vec{\Omega}}{\lambda c}, \tag{1}$$

where $N$ is the number of turns of one loop of the beams, $\vec{A}$ is the area vector of the closed loop and its direction is perpendicular to the plane of the loop, $\vec{\Omega}$ is the angular velocity of the system, $\lambda$ is the laser wavelength and $c$ is the speed of light.

Equation (1) is valid for any arbitrary loop as shown in [18]. Moreover, it matters whether the area that is circumnavigated by the interferometer loop is at an angle or is perpendicular to the axis of rotation. Hence, the phase shift is also proportional to the projection of the area onto the plane that is perpendicular to the axis of rotation. This projection rule is a reminiscent of the mechanism that also applies to the Foucault pendulum.

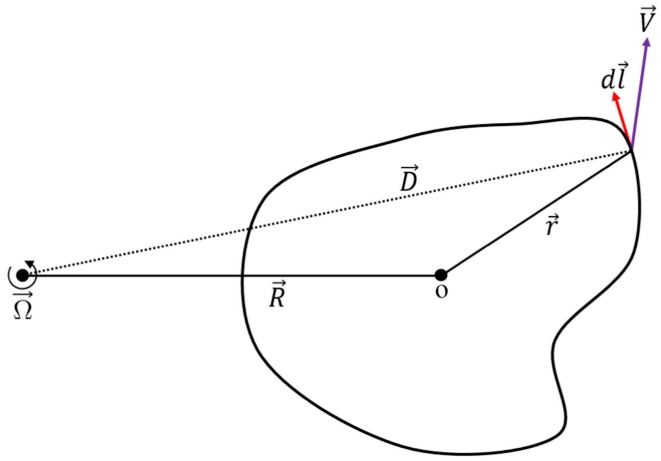



**Figure 1.** Enclosed arbitrary shape loop where the rotation axis is at a distance $\vec{R}$ from the principal area axis of the loop at point o.

Intriguingly, the motion of a Foucault pendulum can also be thought of as a beat note in a superposition of clockwise and counterclockwise swing of a conical pendulum.

The derivation of equation (1) originates in the fact that there is a non-trivial differential time difference ($\mathrm{d}\Delta t$) between the two counter–propagating beams in the closed loop. For small rotation velocities (compared to the speed of light) this is known to be,

$$\mathrm{d}\Delta t = \frac{2\mathrm{d}\vec{l}\cdot\vec{V}}{c^2}, \tag{2}$$

where $\mathrm{d}\vec{l}$ is a line element on the loop, moving at a velocity $\vec{V}$ (see figure 1) [18, 19].

As shown in figure 1, in the case where the rotation axis, $\vec{\Omega}$, is located at the distance $\vec{R}$ from the principal axis of the enclosed loop (o), then the velocity of the line element $\mathrm{d}\vec{l}$ will be $\vec{V} = \vec{\Omega}\times\vec{D}$. Therefore, integration of equation (2) over the closed path multiplied by $N$ turns gives,

$$\Delta t = N\oint\frac{2\mathrm{d}\vec{l}\cdot(\vec{\Omega}\times\vec{D})}{c^2}. \tag{3}$$

The phase difference between the beams is $\Delta\phi = 2\pi\Delta t c/\lambda$, therefore by using the geometrical relation $\vec{D} = \vec{r} + \vec{R}$ we get,

$$\Delta\phi = N\oint\frac{2\mathrm{d}\vec{l}\cdot(\vec{\Omega}\times\vec{r})}{\lambda c} + N\oint\frac{2\mathrm{d}\vec{l}\cdot(\vec{\Omega}\times\vec{R})}{\lambda c}. \tag{4}$$

The first term on the right hand side gives equation (1) as shown in [12]. The second term on the right hand side is a closed line integral over a constant vector, $\vec{\Omega}\times\vec{R}$, and therefore, according to Stokes theorem ($\vec{\nabla}\times(\vec{\Omega}\times\vec{R}) = 0$), equals zero. We thus get that the Sagnac effect is indifferent to the rotation axis position, i.e. the effect is independent of $\vec{R}$.

The theoretical derivation thus far was for light traveling in vacuum. In reality, the two counter–propagating beams will travel through a guiding medium (optical fiber for example) with refractive index $n$ which involves relativistic calculations. However, it has previously been shown that the phase difference is independent from the specific material used to

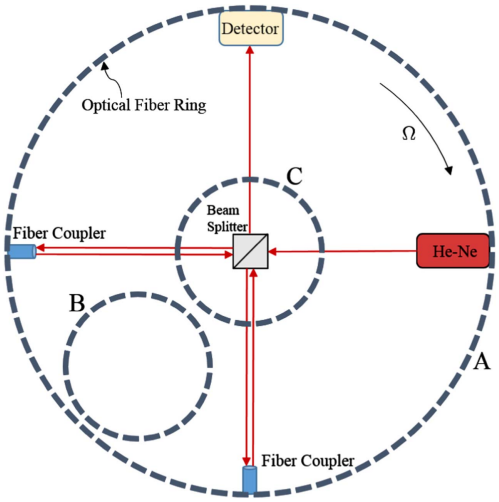



**Figure 2.** Schematics of the fiber ring Sagnac apparatus. (A) Large circumference OFR—center rotation axis. (B) Small circumference OFR—center rotation axis. (C) Small circumference OFR—off-center rotation axis.

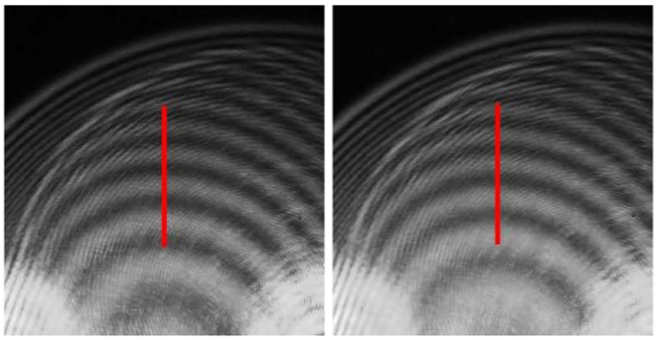


**Figure 3.** Interference pattern at $\Omega = 0$ rad s$^{-1}$ (left) and $\Omega = 2.75$ rad s$^{-1}$ (right). The red line represents the trajectory of the intensity profile of the interference pattern analyzed.

conduct the light beam, and is identical to the vacuum expression given in equation (1) [19, 20].

## 3. Experiment

The laboratory setup for this experiment employs a simple, commonly used, OFR as a Sagnac interferometer. Starting with a single HeNe laser (632 nm) as a light source, a beam splitter divides the light into two beams of approximately identical intensity. Then, each beam is coupled into opposite ends of the OFR and travels along an identical common optical path, but in opposite directions. The system is constructed such that the beams recombine at the beam splitter and interfere. The resulting output beam (at the detector, see figure 2) shows an interference pattern that shifts in proportion to the angular velocity, as shown in figure 3, for

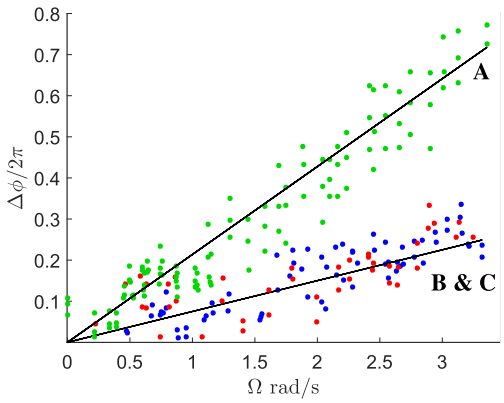



**Figure 4.** Results for the accumulated phase ($\Delta\phi$ normalized by $2\pi$) versus angular velocity ($\Omega$) for the different cases. (A) (green) On-axis large ring interferometer, $D = 115 \pm 0.2$ cm, $N = 9.75 \pm 0.05$. (B) and (C) Off–center axis (blue) and on–center axis (red) small ring interferometer, $D = 35.7 \pm 0.2$ cm, $N = 35.5 \pm 0.05$. Black lines represents the theoretical prediction for both cases according to equation (1).

cases of no rotation (left) and 2.75 rad $s^{-1}$ angular velocity (right). The table upon which the system is constructed can rotate about its central axis—and its rotation rate can be measured by a readout of an optical pattern (a Mayler strip that is connected to a circular rail under the table is sampled by an optical detector which is read by an oscilloscope—so the frequency of the voltage signal can be translated into angular frequency).

Three scenarios of fiber rings were investigated as shown previously in figure 2. (A) The maximum diameter of the round table where the rotation axis was centered in the middle of the fiber ring. (B) The same fiber as before, only with smaller circumference, (thus more loops), positioned in a way such that the rotation axis was off-centered from the middle of the fiber ring. (C) Repeating the setup in (B) with the repositioning of the fiber ring center to coincide with the rotation axis of the table.

The idea behind the data analysis is that since (to a good approximation) the location of the fringes is linearly related to the phase between the beams [18], we can analyze the interference patterns obtained at various rotation velocities and infer the phase difference between the beams from the relative shift of their fringes ($\Delta x_{\text{fringe}}$). This enables us to plot phase difference versus the angular velocity and compare its behavior with our theoretical model.

In figure 3, as an example, an intensity profile of the interference pattern is taken along the trajectory shown as a red line. The interference pattern is not linear, contrary to Young's double slit experiment, hence the phase shift per fringe depends also on the fringe width. The circular shape of the interference patterns creates a width difference between fringes that are closer to the center and fringes that are further out. Therefore, for every angular frequency an interference pattern was recorded, and the phase difference, $\Delta\phi_{\text{exp}}$, was calculated by measuring the relative dark fringe shift, $\Delta x_{\text{fringe}}$, and scaled by the estimated width of the fringe, $\delta$, through the relation,

$$\Delta\phi_{\text{exp}} = 2\pi\frac{\Delta x_{\text{fringe}}}{\delta}. \tag{5}$$

For the case of the large circumference ring (in figure 2(A)), the OFR diameter was $D = 115 \pm 0.2$ cm and the number of turns of the fiber was $N = 9.75 \pm 0.05$. For the small OFRs (in figure 2(B) and (C)), the parameters were $D = 35.7 \pm 0.2$ cm and $N = 35.5 \pm 0.05$. The average fringe width was $\delta = 61$ pixels. Using equation (1) we get the theoretical phase differences per angular frequency to be,

$$\begin{aligned} \Delta\phi^{A}_{\text{Theo}}/\Omega &= 1.344 \pm 0.006 \ \text{rad/rad s}^{-1}, \\ \Delta\phi^{B,C}_{\text{Theo}}/\Omega &= 0.472 \pm 0.011 \ \text{rad/rad s}^{-1}. \end{aligned} \tag{6}$$

The corresponding experimental phase differences per angular frequency were calculated by fitting a linear regression model to the data ($\Delta\phi = a \cdot \Omega + \text{b}$, where 'a' and 'b' are the correlation coefficient and the intercept respectively). For the different data sets, this regression model gave correlation coefficients ('a') and standard errors that are,

$$\begin{aligned} \Delta\phi^{A}_{\text{exp}}/\Omega &= 1.304 \pm 0.038 \ \text{rad/rad s}^{-1}, \\ \Delta\phi^{B}_{\text{exp}}/\Omega &= 0.513 \pm 0.072 \ \text{rad/rad s}^{-1}, \\ \Delta\phi^{C}_{\text{exp}}/\Omega &= 0.495 \pm 0.036 \ \text{rad/rad s}^{-1}. \end{aligned} \tag{7}$$

The comparison between the theoretical and experimental values shows a good agreement (3%–8% error). Figure 4 presents the results for all three cases investigated. The green points are for case (A), blue represents case (B) and red is for case (C). The black lines are the theoretical predictions given in equation (1), for the different experimental parameters used. The phase differences were normalized by $2\pi$. These results clearly show the independent nature of the Sagnac effect from the position of the rotation axis.

## 4. Conclusions

In this work, a simple OFR interferometer was implemented to demonstrate the Sagnac effect, with different diameters and different positions of the ring principal axis. The results confirmed that the interference phase shift is indifferent to the position of the rotation axis relative to the interferometer area principal axis. The experimental results are in very good agreement with the theoretical predictions calculated for the different parameters. Furthermore, the linear relation of the phase difference to the rotation frequency was re-confirmed.

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